

Lab Problem 7

Objective:

Implement Gaussian Discriminant Analysis (GDA) using the Iris dataset to classify Iris flower species based on Sepal Length and Sepal Width, and visualize the decision boundaries.

Dataset Description:

- Use the **Iris dataset** containing **150 samples** from three classes: **Setosa**, **Versicolor**, and **Virginica**.
- Use the features **Sepal Length** and **Sepal Width** for classification.
- Let **X** denote the feature matrix and **y** denote the class labels.

Tasks:

Task 1: Data Preparation

Load the Iris dataset and extract the first two features (**Sepal Length** and **Sepal Width**). Store them in **X**, and store the class labels in **y**.

Task 2: Parameter Estimation

Let **d** be the number of features, **K** the number of classes, and **n** the number of samples.

Initialize the following parameters:

- **mus** (**K × d**) for class mean vectors
- **Sigmas** (**K × d × d**) for covariance matrices
- **phis** (**K**) for class prior probabilities

For each class **k**, compute the **mean vector**, **covariance matrix**, and **class prior probability**. Print the computed **mean vectors**.

Task 3: Prediction

Implement a function `GDA_predictions(x, mus, Sigmas, phis)` that computes

$$P(y|x) \propto P(x|y)P(y)$$

and predicts the class label using

$$\hat{y} = \arg \max_y P(y|x)$$

Task 4: Visualization

- Predict the classes for all samples. Create a **mesh grid** over the feature space and compute predictions for each grid point.
- Plot the **decision regions**, **Gaussian contour lines**, and **training data points**. Label the axes as **Sepal Length** and **Sepal Width**.